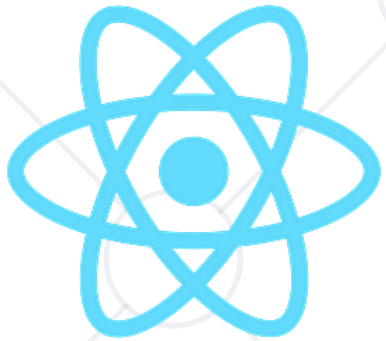
Authentication Build and Deploy



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Development Setup

Environment Overview

- Why can’t we just use Expo Go and skip setup?
  - Because real apps need **more** than previewing JavaScript.
  - Expo Go is a **sandbox**
  - **Limited** native capabilities
  - **Cannot** test production builds
  - **No control** over signing & releases
  - Custom native modules may **fail**
  - Store deployment requires **full toolchain**

Expo Go is for learning and quick iteration — not for shipping.

What Development Setup Gives You

- Professional capability
  - Build **real** binaries (APK / AAB)
  - Install **custom** native dependencies
  - Control environments
  - Debug native errors
  - Prepare for **store submission**

If you want to publish → you need the setup.

Node.js

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- JavaScript runs **outside** the browser
  - **Executes** build tools
  - Runs **Metro bundler**
  - **Installs** dependencies
  - **Powers** scripts
  - **Required** for everything
  - Use **LTS** version

Wrong Node version = weird errors.

Java Development Kit

- Android apps require **Java tooling**
  - Needed for Gradle builds
  - Required by Android SDK
  - Powers native compilation
  - Must match supported versions
  - Set via JAVA_HOME

Without JDK → no Android build.

Android Studio & SDK

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- The Android toolbox
  - Emulator manager
  - SDK installer
  - Platform tools
  - Logcat debugging
  - Device configuration

You don’t need to write Java —but you need its infrastructure.

Android Studio is mission control.

Running on Android Emulator

- Your virtual test device
  - Created via AVD Manager
  - Simulates real hardware
  - Different OS versions
  - No physical phone needed
  - Slower than real devices

```
npm run android
```

Good for development, not perfect for reality.

Running on a Physical Device

- Real hardware, real behavior
  - Enable **Developer Options**
  - Turn on **USB debugging**
  - Connect via **cable** or **Wi-Fi**
  - Better performance metrics
  - Essential before release

Emulators lie. Devices don't.

Environment Variables

- Apps talk to **different** backends
  - Dev / staging / production
  - API URLs change
  - Keys change
  - Never hardcode secrets
  - Use **.env** files

```
API_URL=https://api.myapp.com
```

Configuration is part of architecture.

Build

What Is a Build?

- A build **converts** code into an installable app
  - JavaScript bundled
  - Assets optimized
  - Native binaries created
  - Signed for stores
  - Environment injected

Build = preparation for users.

Native Output Artifacts

- What do we actually produce after a build?
  - Android → APK or AAB
  - iOS → IPA
  - Installable on real devices
  - Uploadable to stores
  - Contains JS + native code + assets

Your app becomes a distributable product.

Development vs Production Builds

- They are very different animals
  - Dev → **fast**, **debug friendly**
  - Prod → **optimized**, **minified**
  - Dev has **warnings**
  - Prod has **performance**
  - Always **test** production builds

Never trust dev behavior.

Local Build

- Build directly on **your machine**
  - Uses installed **SDKs**
  - Full control
  - Useful for debugging
  - Can be slow
  - Setup sensitive

Local builds expose real problems early.

EAS Build

- Cloud builds made **simple**
  - No local native setup
  - iOS + Android
  - Handles certificates
  - Profiles for environments
  - CI/CD friendly

```
eas build --profile production
```

EAS removes 80% of mobile pain.

App Configuration

- Metadata matters
  - App name
  - Bundle identifier
  - Version & build number
  - Icons & splash
  - Permissions

Stores reject sloppy configuration.

App Icons & Splash Screens

- **First impressions** are part of engineering
  - **Required** by stores
  - Multiple sizes needed
  - Platform guidelines apply
  - Generated during build
  - Must match brand identity

Users judge quality in milliseconds.

Versioning

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- Users **upgrade**, not reinstall
  - Increase build number every release
  - Semantic versioning helps
  - Required for store submission
  - Track what changed
  - Vital for debugging

Versioning is communication.

Deployment

What Is Deployment?

- Deployment makes the build **reachable**
  - Upload to store
  - Release to testers
  - Publish updates
  - Rollout gradually
  - Monitor health

Code exists only when users can download it.

Apple App Store

- Strict but premium
  - Manual review
  - Strong **UI** expectations
  - Privacy rules serious
  - Certificates **mandatory**
  - Slower approvals

Quality bar is high.

Google Play

- Faster, **more flexible**
  - Automated checks
  - Closed / open testing tracks
  - **Easier** iteration
  - Still policy heavy
  - **Faster** releases

Speed is your advantage.

Store Assets

- Your marketing page
  - Screenshots
  - Feature graphics
  - App description
  - Keywords
  - Support URL

Users install based on this.

OTA Updates (Over-The-Air)

- Ship **JS updates** without store review
  - Push bug fixes **instantly**
  - Great for **UI tweaks**
  - Cannot change native code
  - Requires **careful** testing
  - Huge productivity boost

Powerful but dangerous if abused.

Distribution

Internal Testing

- Start small
  - Team devices
  - QA validation
  - Quick iterations
  - Catch obvious bugs
  - Validate flows

Never surprise real users.

Beta Testing

- Real users, limited risk
  - Invite lists
  - Feedback loops
  - Crash discovery
  - Performance validation
  - Usability insights

Your best improvements come from here.

Release Strategies

- How **updates** reach users
  - Big bang release
  - Gradual rollout
  - Region based
  - Feature flags
  - Kill switches

Control risk, always.

Monitoring After Release

- Launch is day one
  - Track crashes
  - Monitor performance
  - Watch reviews
  - Measure engagement
  - Patch fast

Shipping starts the real work.

Android Sideloading

- Install your app without the store.
  - Direct APK installation
  - No store review required
  - Useful for internal testing

Great for development. Not for mass users.

Firebase Authentication and Services

Firebase Overview

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- Backend power without backend teams
  - Authentication
  - Firestore database
  - Cloud functions
  - Analytics
  - Push notifications

Firebase accelerates startups.

Authentication

- Handles identity management
  - Secure
  - Battle tested
  - Easy SDK
  - Manages tokens
  - Supports providers

```
signInWithEmailAndPassword(auth, email, pass);
```

Don’t build auth yourself.

Functions

- Backend logic **without managing servers**
  - Run code in the **cloud**
  - Triggered by events or **HTTP**
  - Scales **automatically**
  - Pay for execution
  - Integrates with other **Firebase** services

Write backend power with frontend speed.

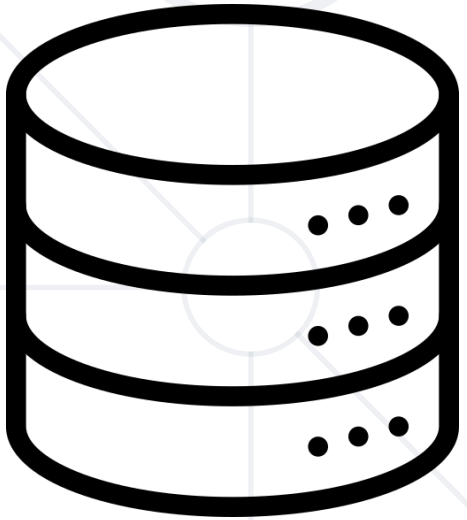
Firestore

- Realtime cloud database
  - NoSQL
  - Realtime listeners
  - Offline support
  - Scales automatically
  - Simple API

Great for rapid products.

Storage

- Store and serve user files from the cloud
  - **Upload** images, videos, documents
  - Built on **Google Cloud** infrastructure
  - Secure via authentication rules
  - Scales **automatically**
  - Designed for **mobile networks**



Perfect for user-generated content.

Summary

- Moving **beyond** Expo Go to a full local or cloud (EAS) environment is essential for custom native modules and store deployment.
- **Reliable builds** require specific versions of Node.js, JDK, and Andr oid Studio to transform JavaScript into installable binaries (APK).
- **Success** depends on proper app configuration, environment variables, and rigorous testing on physical devices rather than just emulators.
- **Publishing** involves managing store assets, utilizing OTA updates for quick fixes, and leveraging Firebase for rapid backend integration (Auth, DB, and Storage).

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